Tutorial 7

EEE301: Signals and Systems
Continuous and Discrete-Time Filters

1. Determine the order for the lowpass filter that fulfil following specification.
   - $\omega_p = 100, \omega_s = 200, \alpha_p = 0.5dB$ and $\alpha_s = 20dB$

2. Design a bandpass Butterworth filter transfer function with the following amplitude response specification:
   - $\omega_{p1} = 1000, \omega_{p2} = 2000, \omega_{s1} = 450, \omega_{s2} = 4000,\ \alpha_p = 2.4dB$
   - $A_{\max} \leq 1dB$ and bandstop gain, $\alpha_s \leq 20dB$

3. Design bandstop Butterworth filter transfer function with the following amplitude response specifications;
   - $\omega_{p1} = 60, \omega_{p2} = 260, \omega_{s1} = 100,\ \omega_{s2} = 150$
   - $\alpha_p = 2.2dB$ and bandstop gain, $\alpha_s = 0.1\ (20dB)$.

4. Parameters for a low pass Chebyshev filter are;
   - Maximum weakening of lowpass is 0.5dB
   - Bandstop weakening at least 22dB at $\omega = 2.33rad/s$.
   - $\omega_p = 1rad/s$

   Find the value of ε, n, and ω_c.

5. Design a bandpass Chebyshev filter transfer function with the following amplitude response;
   - $\omega_{p1} = 1000, \omega_{p2} = 2000, \omega_{s1} = 450, \omega_{s2} = 4000$
   - $A_{max} \leq 1dB$ and bandstop gain, $\alpha_s \leq 20dB$.

6. Draw the direct and cascade form of the FIR filter with the following transfer function
   $$H(z) = 4(1 + 0.35z^{-1} + 0.25z^{-2} + 0.665z^{-3} - 0.45z^{-4})$$

7. Determine the direct form for the FIR filter given by the transfer function
   $$H(z) = 1 + z^{-1} - 0.5z^{-2} + 0.02z^{-3} + 0.003z^{-5}.$$

8. Given a transfer function
   $H(z) = \frac{1 + 0.1z^{-1}}{(1 + 0.3z^{-1})(1 + 0.5z^{-1})}$,
   obtain the direct form structures to realize it.